

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
import os

import cv2
import numpy as np
import matplotlib.pyplot as plt

print("cv2:", cv2.__version__)
print("Libraries loaded successfully!")

cv2: 5.0.0
Libraries loaded successfully!

image_path = "imagee.jpg"

image = cv2.imread(image_path)

if image is None:
    print("ERROR: Image not found!")
else:
    print("Image loaded successfully!")
    print("Image shape:", image.shape)

Image loaded successfully!
Image shape: (512, 600, 3)

# =====
# 3. IMAGE PROPERTIES
# =====

height, width, channels = image.shape

print("\n===== IMAGE PROPERTIES =====")
print("Width      :", width)
print("Height     :", height)
print("Channels    :", channels)
print("Resolution  :", width, "x", height)
print("Data Type   :", image.dtype)
print("Total Pixels:", image.size)

===== IMAGE PROPERTIES =====
Width      : 600
Height     : 512
Channels    : 3
Resolution : 600 x 512
```

Data Type : uint8  
Total Pixels: 921600

```
# =====  
# 4. SAVE IMAGE IN JPEG AND PNG  
# =====
```

```
cv2.imwrite("output_image.jpg", image)  
cv2.imwrite("output_image.png", image)
```

```
print("\nImages saved successfully:")  
print("output_image.jpg")  
print("output_image.png")
```

Images saved successfully:  
output\_image.jpg  
output\_image.png

```
# 5. CONVERT TO GRAYSCALE  
# =====
```

```
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)  
cv2.imwrite("grayscale.jpg", gray)
```

True

```
# 6. CONVERT TO HSV  
# =====
```

```
hsv = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)  
cv2.imwrite("hsv.jpg", hsv)
```

True

```
# 7. CONVERT TO LAB  
# =====
```

```
lab = cv2.cvtColor(image, cv2.COLOR_BGR2LAB)  
cv2.imwrite("lab.jpg", lab)
```

True

```
# 8. RESIZE IMAGE  
# =====
```

```
new_width = 400  
new_height = 300
```

```
resized = cv2.resize(image, (new_width, new_height))
cv2.imwrite("resized.jpg", resized)
```

True

```
# 9. ROTATE IMAGE
```

```
# =====
```

```
# Rotate 90 degrees clockwise
```

```
rotated = cv2.rotate(image, cv2.ROTATE_90_CLOCKWISE)
```

```
cv2.imwrite("rotated.jpg", rotated)
```

True

```
# 10. HORIZONTAL FLIP
```

```
# =====
```

```
horizontal_flip = cv2.flip(image, 1)
```

```
cv2.imwrite("horizontal_flip.jpg", horizontal_flip)
```

True

```
# 11. VERTICAL FLIP
```

```
# =====
```

```
vertical_flip = cv2.flip(image, 0)
```

```
cv2.imwrite("vertical_flip.jpg", vertical_flip)
```

True

```
# 12. IMAGE COMPLEMENT / NEGATIVE
```

```
# =====
```

```
negative = 255 - image
```

```
cv2.imwrite("negative.jpg", negative)
```

True

```
# 13. CROP REGION OF INTEREST (ROI)
```

```
# =====
```

```
# Get image dimensions
```

```
h, w = image.shape[:2]
```

```
# Define ROI
```

```
# Format: image[y1:y2, x1:x2]
```

```

x1 = int(w * 0.25)
x2 = int(w * 0.75)

y1 = int(h * 0.25)
y2 = int(h * 0.75)

roi = image[y1:y2, x1:x2]

cv2.imwrite("roi.jpg", roi)

True

# 14. DISPLAY ALL PROCESSED IMAGES
# =====

plt.figure(figsize=(14, 10))

# Original
plt.subplot(3, 3, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original")
plt.axis("off")

# Grayscale
plt.subplot(3, 3, 2)
plt.imshow(gray, cmap="gray")
plt.title("Grayscale")
plt.axis("off")

# HSV
plt.subplot(3, 3, 3)
plt.imshow(cv2.cvtColor(hsv, cv2.COLOR_HSV2RGB))
plt.title("HSV")
plt.axis("off")

# LAB
plt.subplot(3, 3, 4)
plt.imshow(cv2.cvtColor(lab, cv2.COLOR_LAB2RGB))
plt.title("LAB")
plt.axis("off")

# Resized
plt.subplot(3, 3, 5)
plt.imshow(cv2.cvtColor(resized, cv2.COLOR_BGR2RGB))
plt.title("Resized")
plt.axis("off")

# Rotated
plt.subplot(3, 3, 6)
plt.imshow(cv2.cvtColor(rotated, cv2.COLOR_BGR2RGB))
plt.title("Rotated")

```

```
plt.axis("off")

# Horizontal Flip
plt.subplot(3, 3, 7)
plt.imshow(cv2.cvtColor(horizontal_flip, cv2.COLOR_BGR2RGB))
plt.title("Horizontal Flip")
plt.axis("off")

# Negative
plt.subplot(3, 3, 8)
plt.imshow(cv2.cvtColor(negative, cv2.COLOR_BGR2RGB))
plt.title("Negative")
plt.axis("off")

# ROI
plt.subplot(3, 3, 9)
plt.imshow(cv2.cvtColor(roi, cv2.COLOR_BGR2RGB))
plt.title("ROI")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Original



Grayscale



HSV



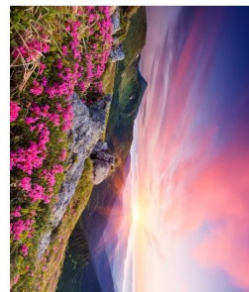
LAB



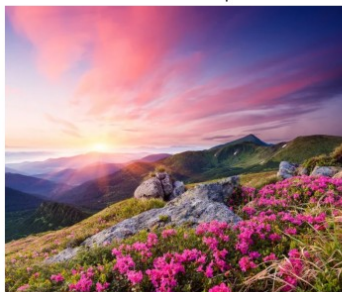
Resized



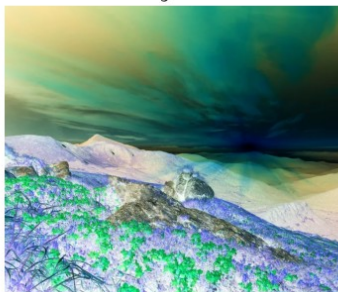
Rotated



Horizontal Flip



Negative



ROI

